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SUPERSEDING
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FEDERAL STANDARD STOCK CATALOG

Section IV

(Part 5)

FEDERAL SPECIFICATION

FOR

MATTRESSES; COTTON (FELTED)

This specification was approved on the above date by the Director of Procurement, for the use of all departments and establishments of the Government and shall become effective not later than May 15, 1940. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATIONS.

A–1. The following Federal specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

V–T–276—Thread; Cotton.
CCC–T–191—Textiles; General Specifications, Test Methods.
CCC–T–351—Ticking; Mattress and Pillow.
DDD–S–751—Stitches; Seams and Stitching.

B. TYPES AND GRADES.

B–1. *Types.*—Mattresses in accordance with this specification shall be furnished in the following types:

Type I. Mattresses with plain box edge.
Type II. Mattresses with rolled box edge.
Type III. Mattresses with plain edge, not boxed.

B–2. *Grades.*—Mattresses in accordance with this specification shall be furnished in the following grades:

Grade A. Mattresses with linters and cotton felt.
Grade B. Mattresses with all linters felt.

C. MATERIAL AND WORKMANSHIP.

C–1. *Materials.*—Shall be as follows:

C–1a. *Linters.*—Shall be new and not lower in grade than United States Standard Subgrade No. 2 high. (See U. S. Department of Agriculture Service and Regulatory Announcement No. 94 and U. S. Department of Agriculture Miscellaneous Publication No. 242.)

C–1b. *Cotton.*—Shall be new, have a staple length of not less than ⅞-inch and be not lower in grade than Low Middling, United States

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Standard. (See U. S. Department of Agriculture Service and Regulatory Announcements Nos. 117 and 150.)

C–1c. *Grade A felt.*—Shall be equal to or better than felt made from a blend of:

Not more than 75 percent linters as specified in C–1a.
Not less than 25 percent cotton as specified in C–1b.

The blend shall be thoroughly dusted and well garnetted so as to produce unbroken laminated sheets of interlaced garnetted felt equal in loftiness to the U. S. Navy Standard sample of Grade B felt. (See par. I–3.)

C–1d. *Grade B felt.*—Shall be made from linters (cotton may be added if desired, the addition of cotton is not mandatory.) The linters shall be thoroughly dusted and well garnetted so as to produce unbroken laminated sheets of interlaced garnetted felt equal to the U. S. Navy Standard sample of all linters felt. (See par. I–3.)

C–1e. *Ticking.*—Shall be type II ticking as covered by Federal Specification CCC–T–351, except that widths shall be as provided by paragraph D–1 hereof.

C–1f. *Thread, sewing.*—Shall be 30/3 cord, type IA1, unbleached cotton thread as covered by Federal Specification V–T–276.

C–1g. *Twine (side, hand stitching).*—Shall be either a flax or hemp twine having a tensile strength of not less than 29 pounds, single-strand test.

C–1h. *Twine (roll-edge stitching).*—Shall be flax twine having a tensile strength of not less than 24 pounds, single-strand test.

C–1i. *Tufting twine or tape.*—Shall have a tensile strength of not less than 32 pounds unless a single ⅝-inch tape is used, in which case the strength shall be not less than 50 pounds.

C–2. *Workmanship.*—Shall be first class in every respect. The cotton felt shall be distributed in layers throughout the mattresses so as to produce a uniform thickness, and tufts and side stitches shall be evenly spaced and drawn. The finished mattress shall be free from avoidable imperfections of manufacture and from defects affecting the appearance or that might affect its serviceability.

D. GENERAL REQUIREMENTS.

D–1. *Case, mattress ticking.*—The top and bottom sections of the mattress shall each be made up of not more than 2 longitudinal pieces, properly matched and sewn together. Cross seams are prohibited. The distance between the seams forming the box edge shall be 2 inches (plus or minus ½ inch) less than the thickness of the mattress, using the method prescribed by paragraph F–4. The box edge, if separate, shall be formed of one piece.

D–2. *Stitches, Seams, and Stitching.*—

D–2a. *Seams and Stitch Types.*—Shall conform to the requirements of Federal Specification DDD–S–751.

D–2b. *Seams.*—The ticking shall be seamed with seam type L–Sb–2 or SSa–1. The closing end shall be seamed with seam type SSc–1.

D–2c. *Stitching.*—All machine stitching for seaming the ticking shall be with stitch type 301 or 401 using 30/3 cord unbleached cotton thread with not less than 10 stitches per inch.

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D–3. *Tufting.*—Mattresses shall be well tufted on both sides with either twine, tape tying tufts or tape with corrosion-resistant metal locking strip, securely fastened to each end. The tying tufts shall be drawn through both sides of the mattress and knotted. Tufts shall be drawn on both sides to a length not more than the thickness of the mattress, using the method prescribed by paragraph F–4, minus 2 inches. In passing through the ticking the up and down twine or tape shall be sufficiently far apart to hold securely and avoid ripping and tearing of the intervening ticking. (Eyelets or cotton tufts will be cause for rejection.) Tufts shall be given diamond spacing and shall be not more than 8 inches apart.

D–4. *Labels.*—Each mattress shall have a label attached to conform with the legal requirements of the state in which the mattress is manufactured. Each mattress also shall have a contractor's cloth label to comply with the requirements of each procurement agency.

D–5. *Standard samples.*—When so specified, the finished mattresses shall conform in quality, workmanship, and appearance to the standard samples established by the procurement agency concerned.

E. DETAIL REQUIREMENTS.

E–1. *Sizes and weights.*—Sizes and weights, unless otherwise specified, shall conform to the requirements of table I. Other sizes, when required, shall be of the length, width, and weight stated in the invitation for bids, within the tolerances given in table I.

TABLE I

Size	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Length (inches), plus or minus ½ inch	76	75	77	78	76	75	76	76	78	78	82	78	73	78
Width (inches), plus or minus ½ inch	26	29	28	31	32	33	36	34	36	36	36	39	53	54
Weight (minimum), pounds	14	17	17	23	23	23	26½	23½	28½	35	37	38	47½	52

E–2. *Stitching.*—

E–2a. *Two-row finishing stitches.*—Type I mattresses shall have not less than the number of stitches given in table II. When sizes not tabulated are specified, the number of stitches shall be interpolated from table II.

TABLE II

Size	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Minimum stitches per single row:														
On each side	(1)	49	50	51	49	49	49	49	51	51	53	51	47	51
On each end	(1)	22	20	22	23	22	26	24	26	26	26	28	38	38

[1] Type III only.

E–2b. *Roll edge stitching (type II mattress).*—Shall be stitched with not less than 2 stitches to every 3 inches.

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F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Sampling of finished mattress.*—One mattress of each grade offered for delivery shall be taken at random by the inspector and forwarded to the point where inspection will be made.

F–2. *Factory inspection.*—Raw materials in process of manufacture and the finished mattress may be subject to preliminary factory inspection at the point of production. Final inspection of the finished mattress shall be made either at the point of production or at a point of delivery, as designated by the procuring agency. The contractor shall inspect all finished mattresses prior to delivery for final inspection.

F–3. *Felt.*—

F–3a. The cotton and/or linters may be examined for compliance with paragraph C–1a by classifiers licensed by the United States Department of Agriculture, or representative samples, each weighing not less than ½ pound, of the cotton and/or linters, and of the felt, may be forwarded to the administrative office concerned, at Washington, D. C., where inspection by the proper board of the United States Department of Agriculture will be obtained. The opinion of this board as to the true grade or quality will be final.

F–4. *Measurement of thickness.* (*See pars. D–1 and D–3.*)—Thickness shall be taken as the average of measurement taken at each of the four corners of a mattress, made by determining the free height under the corner of an 18-inch square wood panel weighing 4 pounds, and resting on the mattress, with two edges over the adjacent edges of the mattress.

F–5. *Measurement of length and width.*—Length and width shall be taken as the average of three measurements in each direction. Before measuring, mattresses shall be shaken out after removal from cartons or bales, and laid in a flat position.

F–6. *Tufting twine or tape.*—The strength of the tufting twine or tape shall be determined on a standard testing machine, with an initial distance of not less than 10 inches between centers of drums, and a rate of travel of approximately 12 inches per minute for the moving drum.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G–1. *Packing.*—The mattresses shall be packed in individual solid fiber-board cartons, or in bales, without compression, containing 5 mattresses each. The method of packing shall be as specified by the procurement agency in the invitation for bids.

G–1a. *Cartons.*—Shall be made of 0.080–300 points per square inch "Mullen-test" solid fiber board. They shall be of one-piece construction and shall meet all other requirements regarding quality, strength, and design, of specifications, of issue in effect on date of shipment, of Consolidated Freight Classification. Cartons shall be dustproof when closed and sealed.

G–1b. *Bales.*—Five mattresses shall be packed full length in a non-pressed bale without banding. The outside covering of the bale shall be made of burlap weighing 12 ounces per yard (40 inches). The bale shall be sewed up with three-ply linen or other suitable twine of not less than 40 pounds tensile strength, stitches not less than 1½ inches in length, each stitch to be knotted. Five inches of burlap shall be gathered together forming two "ears" on each end of the bale for

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handles. This is accomplished by turning inward the corners of the burlap, sewing the ears through and through, and passing the twine around the ears close to the bale and continuing the stitch out to the end of the ear. All ends of twine shall be securely fastened by knotting. Each bale shall be interlined with No. 2 kraft paper, 60 pounds per thousand sheets, 24 inches by 36 inches, and shall have underneath the burlap a lining of a double thickness of paraffin-impregnated kraft paper, creped in both directions. The creped paper shall have a tensile strength of not less than 12 pounds per inch and shall stretch not less than 30 percent.

G–2. *Marking.*—Each carton or bale shall be plainly marked on each end with the name of the article, the type, the grade, the size and the quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order. All bales shall be plainly marked "Use no hooks".

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS.

H–1. The following departmental specifications, of the issue in effect on date of invitation for bids, respectively form a part of this specification:

H–2. *Army purchases.*—U. S. Army Specification No. 100–2 Standard Specification for Marking Shipments.

H–2a. *Sampling.*—Samples of any materials, components, etc., not furnished by the United States, entering into the manufacture of the articles covered herein, shall be selected from time to time by the Government inspector and carefully examined and tests made to determine if they are in accordance with this specification.

H–2b. *Inspection.*—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction of material shall not relieve the contractor of responsibility for faulty workmanship or material which may be discovered at any time prior to final acceptance. Final inspection of the finished articles shall be made either at the point of production or at the point of delivery in the contract or purchase order of the procuring agency. In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

H–3. *Navy Department purchases.*—For administrative reasons, Navy Department purchases will be made under the issue in effect on date of invitation for bids of Navy Department Specification 27M9, Mattresses, Cotton (Felted), Berth (Chief Petty Officers' and Sick Bay). This Navy Department specification conforms in detail to the technical requirements of the Federal specification.

H–4. *Veterans' Administration purchases.*—

H–4a. *Workmanship.*—Shall be first class in every respect. The cotton felt shall be distributed in layers throughout the mattresses so as to produce a crown in the center, allowing the thickness of the mattresses to be one inch greater in the center than the thickness at the sides. The tufts and side stitches shall be evenly spaced. The finished mattresses shall be free from avoidable imperfections of manufacture and from defects affecting the appearance or that might affect their serviceability.

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I. NOTES.

I–1. Requisitions, schedules, proposals, contracts, and orders should exercise any desired options specified herein. (See pars. B–1, B–2, D–5, and G–1.)

I–2. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I–3. U. S. Navy standard samples of felt referred to in paragraphs C–1c and C–1d may be seen or portions thereof obtained upon application to the Commandant, Navy Yard, Philadelphia, Pa.

I–4. This specification covers only the types, classes, grades, sizes, etc., of the commodity as generally purchased by the Federal Government, and is not intended to include all of the types, etc., which are commercially available.

I–5. An index of Federal specifications may be purchased as noted in the paragraph next below, price to be obtained from the Superintendent of Documents.

I–6. Copies of this specification, and of the specifications referred to in section A may be obtained upon application, accompanied by money order or coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington, D. C. Price, 5 cents each except DDD–S–751, 25 cents.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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